



First spinning wheel

- **What?** Hand-turned spinning wheel
- **Who?** Unknown
- **Where and when?** Probably India, c. 600 CE

Before materials such as cotton or wool can be woven they need to be spun into threads. Early people used to do this by pulling out and twisting the fibers of the raw material between their fingers. This time-consuming task was greatly speeded up by the invention of the spinning wheel. The hand-turned wheel twists fibers into thread, which it winds around a stick called a “spindle.”

Vertical-axis mill

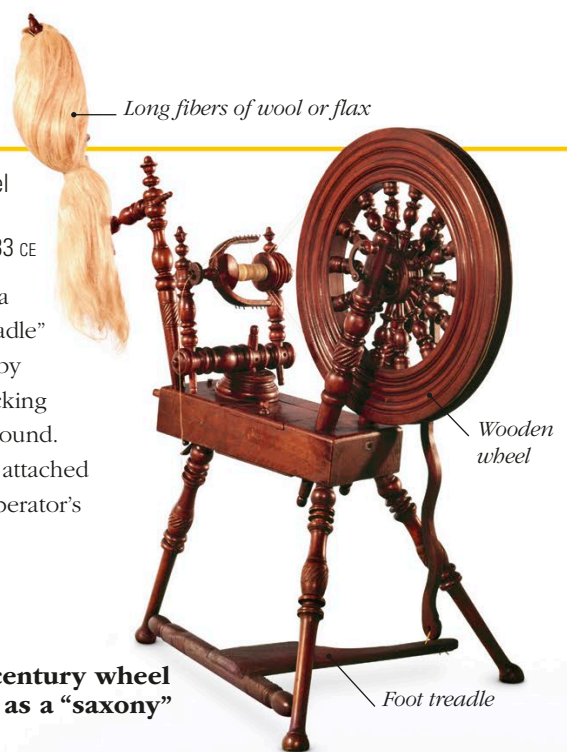
- **What?** Post mill
- **Who?** Unknown
- **Where and when?** Northern Europe, c. 1200 CE

The most common use of windmills was to grind grain into flour, but they were also used for other purposes, such as pumping water. As the sails of a windmill turn in the wind, gears inside the mill use the rotational force to move mechanical parts. A post, or vertical-axis, mill had a large central shaft that allowed the sails to be moved to face the wind.

Treadle wheel

- **What?** Foot-powered spinning wheel
- **Who?** Unknown
- **Where and when?** Germany, c. 1533 CE

As the spinning wheel developed, a foot-operated board called the “treadle” was added. Pressed up and down by the operator’s feet, the treadle’s rocking motion drove the spinning wheel round. The raw fibers were held on a rod attached to the treadle wheel, leaving the operator’s hands free to guide the thread.



An 18th-century wheel known as a “saxony”

▼ WIND POWER

These types of windmill could not be used for industry because they did not produce as much energy as coal-powered engines.

